



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Linear Search -Sort

LINEAR SEARCH -SORT

Read n book page counts into an array. Sort ascending using any method. Use linear search to find the first book having ≥ 300 pages and the last book having ≤ 150 pages. Display page counts and indices (0-based), or "Not found". Validate sorted order; $n \leq 50$.

Sample Input: 7

120 450 275 310 140 500 200

Sample Output (after sort): 120 140 200 275 310 450 500

First ≥ 300 : index 4 (310)

Last ≤ 150 : index 1 (140)

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int a[50];
8     for (int i = 0; i < n; i++) scanf("%d", &a[i]);
9
10
11     for (int i = 0; i < n - 1; i++) {
12         for (int j = 0; j < n - 1 - i; j++) {
13             if (a[j] > a[j + 1]) {
14                 int t = a[j];
15                 a[j] = a[j + 1];
16                 a[j + 1] = t;
17             }
18         }
19     }
20
21
```

OUTPUT

```
0 0 0 120 32759 2085613624 2085613624
First >=300: index 4 (32759)
Last <=150: index 3 (120)
```

INPUT

7

120n450 275 310 140 500 200

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